TECHNISCHE UNIVERSITÄT DARMSTADT

Fachbereich Mathematik

Bachelorarbeit

Representational Measurement of Similarity: The Additive Difference Model, Revisited

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Chapter 1

Introduction

1.1 Proximity Structures and Metrics

In diesem Kapitel geht es erstmal darum die Mathematische Basis Aufzubauen. Wie aus der Einleitung hervorgeht, wollen wir jeweils zwei Paare von Stimuli auf ihre Ähnlichkeit vergleichen. Seien a, b, c und d solche Stimuli. Dann bedeutet die Notation $(a, b) \succsim (c, d)$, dass Stimuli a und b sich unähnlicher sind, als c und d. Analog bedeutet $\sim$, dass die Ähnlichkeit zweier Paare gleich ist.
Die Idee von einer proximity structure, ist die Mathematische Definition dieser Idee: eine Relation $\succsim$ auf einer Menge A die die Grundeigenschaften einer Metrik erfüllt.

Definition 1.1.1 (Factorial Proximity Structure)**.**
Suppose A is a set and $\succsim$ a quaternary relation on A, i.e., binary relation on $A \times A$. $\langle A, \succsim \rangle$ is a proximity structure iff the following axioms hold for all $a, b \in A$:

1. $\succsim$ is a weak ordering on $A \times A$, i.e., it is connected and transitive.
2. $(a, b) \succsim (a, a)$ whenever $a \neq b$.
3. $(a, a) \sim (b, b)$ (minimality).
4. $(a, b) \sim (b, a)$ (symmetry).

The structure is factorial iff
$$A = \prod_{i=1}^{n} A_i.$$

Remark 1.1.2 (Connectedness). Connected means, that comparison of all distinct elements is possible, i.e., for a relation R on a set X it hold for all $x, y \in X$ with $x \neq y$ that either xRy or yRx.

Da wir die Relation $\succsim$ durch eine Metrik repräsentieren wollen, lohnt es sich einen Begriff dafür einzuführen. Die Idee ist das, was man sich intuitiv vorstellt: man will dass die Metrik die Struktur von $\langle A, \succsim \rangle$ erhält, und die einzige Struktur die wir gegeben haben, ist die Ordnung jeweils zweier Paare.

Definition 1.1.3 (Representability of $\succsim$ by a Metric)**.**
A proximity structure $\langle A, \succsim \rangle$ is representable by a metric iff there exists a real-valued function δ on A such that:

1. $\langle A, \delta \rangle$ is a metric space, i.e., δ is a metric on A.
2. δ preserves the relation $\succsim$, i.e., for all $a, b, c, d \in A$
$$\delta(a, b) \geq \delta(c, d) \iff (a, b) \succsim (c, d).$$

Chapter 2

Additive Difference Models and Structures

2.1 Additive Difference Model

Definition 2.1.1 (Additive Difference Model).

1. *Decomposability*: The distance between stimuli is a function of componentwise contributions.

$$\delta(a,b) = F[\psi_1(a_1,b_1),\dots,\psi_n(a_n,b_n)], \tag{2.1}$$

where F is an increasing [1] real-valued function in each of its n real arguments and ψ_i is a real-valued symmetric function satisfying $\psi_i(a_i,b_i) > \psi_i(a_i,a_i)$ when $a_i \neq b_i$ for all $i = 1,...,n$.

2. *Intradimensional subtractivity*: Each componentwise contribution is the absolute value of an appropriate scale difference.

$$\delta(a,b) = F[|\varphi_1(a_1)-\varphi_1(b_1)|,\dots,|\varphi_n(a_n)-\varphi_n(b_n)|], \tag{2.2}$$

where F is as in eq. (2.1) and ψ_i is replaced with $|\varphi_i(a_i)-\varphi_i(b_i)|$ for some real-valued function φ_i defined on A_i for $i = 1,...,n$.

3. *Interdimensional additivity*: The distance is a function of the sum of componentwise contributions.

$$\delta(a,b) = G\left[\sum_{i=1}^{n}\psi_i(a_i,b_i)\right], \tag{2.3}$$

[1] We use the terms increasing and decreasing to denote real-valued functions that are strictly monotonically increasing and decreasing, respectively.

where ψ_i are as in eq. (2.1) and G is an increasing function in one argument.

4. *Additive-difference model*: If both, additivity and subtractivity are assumed, we get the additive difference model.

$$\delta(a,b) = G\left\{\sum_{i=1}^{n} F_i[|\varphi_i(a_i) - \varphi_i(b_i)|]\right\}, \tag{2.4}$$

where G and F_i, for $i = 1, \ldots, n$, are all increasing functions in one argument.

2.2 Additive Difference Structure

2.3 Examples

Example 2.3.1 (p-exp Metric Family)**.**
Let the F in the additive difference model (2.4) be given by

$$F(t) = q \cdot (e^{t \cdot r} - 1)^p \qquad q, r > 0, p \geq 1. \tag{2.5}$$

The model defined by this F is a metric.

Proof. First, we will prove positivity and symmetry of this function, and then turn to the triangle inequality which is a result of theorem which provides a generalization of the Mikowski inequality.

Part 1 (Positivity and Symmetry)**.**
Positivity and symmetry are both straight forward.

1. Positivity: $\delta(a, b) = 0$ implies $F(|a_i - b_i|) = 0$ for all $i = 1, \ldots, n$, which is only possible if $|a_i - b_i| \cdot r = 0$, meaning $a_i = b_i$ for all $i = 1, \ldots, n$.
   Conversely, if $a \neq b$, then $a_i \neq b_i$ for at least one i in $1, \ldots n$, which implies $F_i(|a_i - b_i|) > 0$, and thus $\delta(a, b) > 0$.

2. Symmetry: Symmetry follows directly from the fact that $|a_i - b_i| = |b_i - a_i|$ holds.

The triangle inequality follows from Mullholland's theorem (Mulholland, 1949, p. 297).

Theorem 2.3.2 (Mulholland's Inequality)**.**
The inequality

$$f^{-1}\left[\sum_{i=1}^{n} f(|x_i + y_i|)\right] \leq f^{-1}\left[\sum_{i=1}^{n} f(|x_i|)\right] + f^{-1}\left[\sum_{i=1}^{n} f(|y_i|)\right].$$

holds for all $x, y \in \mathbb{R}^n, n = 1, 2, \ldots$ *whenever the following three conditions are satisfied:*

1. f *is continuous, increasing and* $f(0) = 0$.

2. f *is convex on* $[0, \infty)$.

3. $\log f(e^x)$ *is convex for* $x \in (-\infty, \infty)$.

Part 1 (Proof that eq. (2.5) satisfies Mulholland's conditions)**.**
Regarding Point 1: Continuity and $f(0) = 0$ are clear. For monotonicity, note that $e^{rt} - 1$ is increasing, exponentiation is also increasing, and multiplying by $q > 0$ preserves monotonicity. Thus, the composition is also increasing.
Regarding point 2: Similar reasoning applies for convexity. The composition of convex functions is convex, if the outer function is non-decreasing (Thomson et al., 2008, p. 481, Note 207). Since multiplication by a positive constant and exponentiation for exponents greater than 1 is both convex and increasing, the resulting function must be convex.

We now proceed to prove the convexity of $\log F(e^t)$. By logarithmic properties, it holds that

$$\log q(e^{re^x} - 1)^p = \log q + p \cdot \log(e^{re^x} - 1).$$

Arbitrary addition and multiplication by $p \geq 1$ preserve convexity, so it suffices to show the convexity of $\log(e^{re^x} - 1)$.

We prove this by examining the second derivative. If the second derivative is strictly positive, the function is convex (Hug & Weil, 2020, p. 57, Remark 2.13).

Differentiating twice yields

$$\frac{re^{re^x+x}\left(r(-e^x) + e^{re^x} - 1\right)}{\left(e^{re^x} - 1\right)^2} \overset{!}{>} 0.$$

The denominator is always positive and non-zero, so we can ignore it. Similarly, the factor re^{re^x+x} is always positive and can also be omitted. It remains to show that

$$\begin{aligned} -re^x + e^{re^x} - 1 &> 0 \\ \iff e^{re^x} - re^x &> 1 \end{aligned}$$

holds.

This can be shown in two steps. First, we show that the left-hand side is strictly increasing. Second, we prove that its limit as $x \to -\infty$ is 1. Therefore, the inequality holds for all $x \in \mathbb{R}$.

1. For the first point, we show that the derivative is greater than zero. Differentiation yields

   $$re^x \cdot e^{re^x} - re^x = re^x \cdot (e^{re^x} - 1) > 0$$

   since $e^x > 0$.

2.
$$\lim_{x\to-\infty} e^{re^x} - re^x = \lim_{x\to-\infty} e^{re^x} - \lim_{x\to-\infty} re^x = 1 - 0 = 1$$

Thus, convexity is established, and the claim follows from Mulholland's theorem.

□

Remark 2.3.3 (Minkowski metric is part of the p-exp metric family). If we set $q = r^{-p}$ in (2.5), we obtain the Minkowski metric in the limit as $r \to 0$.

Proof. Setting $q = r^{-p}$ yields

$$r^{-p}(e^{rt}-1)^p = \left(\frac{e^{rt}-1}{r}\right)^p.$$

As $r \to 0$, both the numerator and denominator tend to 0 (we can pull the limit inside the exponent since the exponentiation is continuous), therefore we can apply L'Hôpital's rule. Hence

$$\begin{aligned}
\lim_{r\to 0}\left(\frac{e^{rt}-1}{r}\right)^p &= \left(\lim_{r\to 0}\frac{e^{rt}-1}{r}\right)^p && \text{(continuity)}\\
&= \left(\lim_{r\to 0}\frac{t\cdot e^{rt}-1}{1}\right)^p && \text{(L'Hôpital's rule)}\\
&= t^p.
\end{aligned}$$

□

Das ist von Bedeutung, da dies mithilfe des Hauptresultates **??** zeigt, dass die Metriken der p-exp Familie keine Metriken mit additiven Segmenten sein können, obwohl man die l^p Metrik durch Grenzwertbildung aus dieser Familie darstellen kann. Das bedeutet, dass im Grenzwertprozess aus der Eigenschaft "additiv auf den Koordinatenachsen", die Eigenschaft "additiv auf beliebigen Liniensegmenten" wird.

Not every function that can be represented by the additive difference model is a metric.

Chapter 3

Segmental Additivity

3.1 Segmentally Additive Metrics

Da Metriken an sich wenig Struktur bieten, lohnt es sich noch eine Eigenschaft mehr mehr zu fordern.
Diese Eigenschaft ist segmental additivity und beschreibt, dass es zu zwei Punkten immer eine Kurve gibt, die diese verbindet und die Abstände additiv sind, das heißt auf dem Segment die Dreiecksungleichung mit Gleichheit gilt. Für uns reicht es zu wissen, dass wenn b auf dem Segment zwischen a und c liegt, dass dann $\delta(a,b) + \delta(b,c) = \delta(a,c)$ gilt. Für die genaue Definition siehe Anhang Am Ende dieses Abschnittes sind Beispiele für Metriken mit additiven Segmenten angegeben.

Der Folgende Satz liefert eine (bis auf positive Skalierung) eindeutige Metrik die die Relation repräsentiert und zusätzlich zur Ordnung, die Eigenschaft segmental solvability erhält.

Theorem 3.1.1 (Representation Theorem for Segmentally Additive Proximity Structures)**.**
Suppose $\langle A, \succsim \rangle$ is a segmentally additive proximity structure. Then $\langle A, \succsim \rangle$ is representable by a metric, and additionally, for all $a, b, c \in A$:

1. *$\langle abc \rangle \iff \delta(a,b) + \delta(b,c) = \delta(a,c)$.*
2. *If δ' is another metric on A that satisfies the above conditions, then there exists $\alpha > 0$ such that $\delta' = \alpha\delta$ (δ is a ratio scale).*

Definition 3.1.2 (temp)**.**
Todo: Metric, open, convex, stetig ... definieren?

The definitions and statements from this section are taken out from (Busemann, 1955, Chapter 1).

Definition 3.1.3 (Curves and Segments)**.**

1. Let δ be a metric on X. A parametric curve in $\langle X, \delta \rangle$ is a function γ that maps a closed interval $[a, b] \subset \mathbb{R}$ into X and is continuous. The parametric curve γ is said to join $\gamma(a)$ and $\gamma(b)$.

2. Let γ be a parametric curve in $\langle X, \delta \rangle$ with domain $[a, b]$. Its length $\mathrm{L}(\gamma)$ is the supremum of the sums
$$\sum_{i=1}^{n} \delta \left[\gamma(a_{i-1}), \gamma(a_i) \right]$$
over all finite sequences $a_0, a_1, \ldots, a_n$ such that
$$a = a_0 \leq a_1 \leq \cdots \leq a_n = b.$$
If $\mathrm{L}(\gamma) < \infty$, then γ is rectifiable.

3. A rectifiable curve is a segment iff its length is equal to the distance between its endpoints, i.e. $\mathrm{L}(\gamma) = \delta\,(a,\ b)$.

4. For $a \leq x \leq b$ we denote the length of the sub-curve $\gamma(t),\ a \leq t \leq x$ from a to x by $\mathrm{L}_a^x(\gamma)$. This is also called the *arc length* of the curve.

Remark 3.1.4 (Continuity & monotonicity of sub length). For a rectifiable curve $\gamma(t),\ a \leq t \leq b$, the arc length $\mathrm{L}_a^x(\gamma)$ is a continuous, non-decreasing function of x.

Definition 3.1.5 (Metric Space with Additive Segments)**.**
Let δ be a metric on X. $\langle X, \delta \rangle$ is a metric space with additive segments iff for any $x, y \in X$ there is a segment joining them.

3.2 Segmentally Additive Proximity Structures

Da wir diese Additivität aus der proximity structure herleiten wollen, müssen wir Eigenschaften definieren, aus der man die Additivität herleiten kann. Dazu benötigen wir Folgenden Begriff:

Definition 3.2.1 (Ternary Relation on Proximity Structures)**.**
For $a, b, c \in A$, the ternary relation $(a, b, c\rangle$ is said to hold iff for all $a', b', c' \in A$:

1. If $(a, b) \succsim (a', b')$ and $(a', c') \succsim (a, c)$, then $(b', c') \succsim (b, c)$
2. If either of the preceding inequalities is strict, then the resulting inequality is also strict.

Further the relation $\langle abc\rangle$ is said to hold if both $(a, b, c\rangle$ and $(c, b, a\rangle$ hold.

Die Bedingungen, die notwendig sind, um eine Darstellung durch eine Metrik mit additiven Segmenten zu ermöglichen, sind in der folgenden Definition und dem Satz danach zusammengefasst. Der zweite Teil der Definition wird erst später, für das Hauptresultat (Satz **??**) relevant, im Prinzip geht es darum dass man sich Grenzwerte und Cauchy Folgen anschauen kann, also der normale Begriff von Vollständigkeit.

Definition 3.2.2 (Segmentally Additive Proximity Structure)**.**
A proximity structure $\langle A, \succsim\rangle$, with a ternary relation $\langle\rangle$ defined in terms of $\succsim$, is segmentally additive if and only if the following two axioms hold for all $a, c, e, f \in A$:

1. If $(a, c) \succsim (e, f)$, then there exists $b \in A$ such that $(a, b) \sim (e, f)$ and $\langle abc\rangle$. (segmental solvability)
2. If $e \neq f$, then there exist $b_0, \ldots, b_n \in A$, such that $b_0 = a$, $b_n = c$, and $(e, f) \succsim (b_{i-1}, b_i)$ for $i = 1, \ldots, n$. (connectedness)

A segmentally additive proximity structure is complete if and only if the following two axioms hold for all $a, c, e, f \in A$:

3. If $e \neq f$, then there exist $b, d \in A$, with $b \neq d$, such that $(e, f) \succ (b, d)$. (Non-discreteness of distances)
4. Let $a_1, \ldots, a_n, \ldots$ be a sequence of elements of A, such that for all $e, f \in A$, if $e \neq f$, then $(e, f) \succsim (a_i, a_j)$ for all but finitely many pairs (i, j). Then there exists $b \in A$, such that for all $e, f \in A$, if $e \neq f$, then $(e, f) \succsim (a_i, b)$ for all but finitely many i. (Limits for Cauchy sequences)

3.2.1 Examples

Example 3.2.3 (Segmental additivity of the minkowski metric)**.**
For $p \geq 1$ the Minkowski Metric is a metric with additive segments. The segments are all straight lines (i.e. the line segments between two arbitrary points).

Proof. Let $x, z \in \mathbb{R}^n$ and $0 \leq t \leq 1$. Then, for $y = (1-t)x + tz$ (y is on the line segment between z and x, iff y can be represented in this way) it holds that:

$$\begin{aligned}
d(x,y) + d(y,z) &= \left[\sum_{i=1}^{n} |x_i - (1-t)x_i - tz_i|^p\right]^{1/p} + \left[\sum_{i=1}^{n} |(1-t)x_i + tz_i - z_i|^p\right]^{1/p} \\
&= \left[\sum_{i=1}^{n} |x_i - x_i + tx_i - tz_i|^p\right]^{1/p} + \left[\sum_{i=1}^{n} |x_i - tx_i + tz_i - z_i|^p\right]^{1/p} \\
&= t\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} + (1-t)\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} \\
&= d(x,z).
\end{aligned}$$

□

Chapter 4

Combining Segmental Additivity with the AD-Model

4.1 Conditions for AD-Metrics

Theorem 4.1.1 (Necessary and sufficient conditions for additive difference metrics).
*Suppose $\langle A, \succsim \rangle$ is an additive-difference proximity structure such that f is the additive-difference representation from **??**, i.e.,*

$$f(a,b) = \sum_{i=1}^{n} F_i(|\varphi_i(a) - \varphi_i(b)|).$$

Suppose further that the domain of each F_i is a real interval and $F_i(0) = 0$, $i = 1, \ldots, n$ (through transformation, since F_i are interval scales).

For $\langle A, \succsim \rangle$ to be compatible with a metric, and additionally be additive on the coordinate axes,

1. *it is necessary that there exists a continuous, convex function F, such that $F_i(x) = F(t_i x), t_i > 0$, for all x in the domain of F_i, $i = 1, \ldots, n$; and*

2. *sufficient that, in addition (to the conditions in 1.), the function $\log F(e^x)$ is convex.*

Remark 4.1.2 (Additivity on the coordinate axes). Additivity on the coordinate axes means the following: If $a, b, c \in A$ differ only on one factor A_i, then the triangle inequality of the metric holds with equality. In mathematical terms:

$$a_j = b_j = c_j \text{ for all } j \neq i \text{ and } (a,c) \succsim (a,b), (b,c) \Rightarrow \delta(a,b) + \delta(b,c) = \delta(a,c)$$

In the following lemmas, until the proof of this theorem is finished, everything will be assumed as in theorem 4.1.1, unless otherwise stated.

The following remark ties this result closer to the Minkowski metric, and provides a form which the metric must take. This will play an important role in the main result theorem 4.2.1.

Remark 4.1.3 (Form of δ). Examining section 4.1 more closely yields, that $G = F^{-1}$ must hold and the representation of δ takes the form

$$\delta(a, b) = F^{-1}\Big\{\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\}$$

where F is continuous, convex and increasing.
This follows form two arguments. Firstly, incorporating the t_i into the φ_i

$$\begin{aligned} G\Big\{\sum_{i=1}^{n} F_i(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} &= G\Big\{\sum_{i=1}^{n} F(t_i|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} \\ &= G\Big\{\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} \end{aligned}$$

yields the desired form inside the sum.

Secondly, by using $F_i(x) = F_n(t_i x)$ and $F_i(x) = G^{-1}(t_i x)$. The first equation applied for $i = n$ yields $F_n(x) = F_n(t_n x)$ which implies $t_n = 1$. Using this in the second equation for $i = n$ yields $F(x) = F_n(x) = G^{-1}(t_n x) = G^{-1}(x)$. Hence, by taking the inverse on both sides (which is possible due to F and G both being increasing), we get the desired form for δ.

Proof of theorem 4.1.1 1. According to the assumptions of the theorem, the additive difference representaion $f(a, b)$ is compatible with a metric δ on A, i.e.,

$$f(a, b) \leq f(c, d) \iff \delta(a, b) \leq \delta(c, d) \quad \text{for all } a, b, c, d \in A.$$

Therefore, there exists an increasing function G, such that for all $a, b \in A$

$$\delta(a, b) = G(f(a, b)) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right).$$

A possible perspective on this is, that by Todo: ref f is an interval scale and therefore preserves orderings. Since the ordinal ordering of all pairs of elements from A is same for f and δ, there must be an increasing function which transforms one scale into the other and preserves the ordinal orderings.
Assume a, b, and c differ only in the i-th factor and $a|b|c$. Thus, we can use additivity on the i-th coordinate axis for $a, b, c \in A$ which yields

$$G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) + G\left(F_i\left(|\varphi_i(b_i) - \varphi_i(c_i)|\right)\right) = G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(c_i)|\right)\right).$$

Defining $|\varphi_i(a_i) - \varphi_i(b_i)| =: x$ and $|\varphi_i(b_i) - \varphi_i(c_i)| =: y$ yields

$$G[F_i(x)] + G[F_i(y)] = G[F_i(x+y)].$$

Further, define $H_i(x) = G[F_i(x)]$. This reduces the above equation to $H_i(x) + H_i(y) = H_i(x+y)$ for all x, y in the domain of $F_i, 1 \leq i \leq n$. This means, that H_i must be linear and since the values are positive (because we are still observing metrics), the only monotonic solution to this equation is $H_i(x) = t_i x$ for $t_i > 0$. Since G is increasing, it has an inverse G^{-1} satisfying

$$G^{-1}[H_i(x)] = G^{-1}[t_i x] = F_i(x)$$

where F_i is defined. Since G^{-1} is independent of i, the F_i can only differ in range and in the unit of the domain (i.e. the t_i which defines the domain of F_i by adjusting the domain of G^{-1}). Without loss of generality, we can assume that the range of F_n is the largest. Since the range of F_n includes the range of F_i, we can set $F = F_n$ and obtain $F_i(x) = F_n(t_i x) = F(t_i x)$ for $t_i > 0$ and for every x in the domain of F_i. Continuity and convexity will be proven in the next lemmas. □

Lemma 4.1.4.

$$F(\alpha + \beta + \gamma) + F(\gamma) \geq F(\alpha + \gamma) + F(\beta + \gamma).$$

Proof. In the triangle inequality,

$$F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(\beta_i)\right] \geq F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i + \beta_i)\right],$$

let $\alpha_1 = \gamma, \alpha_2 = F^{-1}(F(\alpha+\gamma) - F(\gamma)), \alpha_j = 0$ for $3 \leq j \leq n$ and $\beta_1 = \beta, \beta_j = 0$ for $2 \leq j \leq n$. Hence,

$$\begin{aligned} F(\alpha + \beta + \gamma) &= F\left[F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(\beta_i)\right]\right] \\ &\geq \sum_{i=1}^{n} F(\alpha_i + \beta_i) \qquad \text{(by the triangle inequality)} \\ &= F(\beta + \gamma) + F(\alpha + \gamma) - F(\gamma). \end{aligned}$$

□

Lemma 4.1.5.
F is continuous.

Proof. Since F is increasing and defined on an interval, all discontinuities are gaps (Thomson et al., 2008, p. 335) [1]. Real analysis yields, that real monotonic functions have at most countable many gaps. We will derive a contradiction from this.

Assume $F(\alpha)$ is the lower boundary of a gap, then there exists $\epsilon > 0$ such that for all $\delta > 0$, $F(\alpha + \delta) - F(\alpha) > \epsilon$. However, by lemma 4.1.4, for every $\beta > 0$,

$$F(\alpha + \beta + \delta) - F(\alpha + \beta) \geq F(\alpha + \delta) - F(\alpha) > \epsilon,$$

and so there is a gap at any $\alpha + \beta$, and F would have to be discontinuous everywhere. This is impossible, so F must be continuous. □

Lemma 4.1.6 (Characterization of Convexity)**.**
Suppose F is a strictly increasing function defined on the nonnegative reals. Then F is convex iff for all $\alpha, \beta, \gamma \geq 0$ the inequality from 4.1.4 holds, i.e.,

$$F(\alpha + \beta + \gamma) + F(\gamma) \geq F(\alpha + \gamma) + F(\beta + \gamma).$$

Proof.

Part 1 (⇒)**.**
Suppose, F is convex, and let $p = \alpha/(\alpha + \beta)$, then

$$\begin{aligned} p(\alpha + \beta + \gamma) + (1 - p)\gamma &= \alpha + \gamma \\ (1 - p)(\alpha + \beta + \gamma) + p\gamma &= \beta + \gamma. \end{aligned}$$

Hence, by convexity,

[1] (Thomson et al., 2008) prove the following statement: Let f be a real-valued monotonic function defined on an interval $[a, b]$. Then the set of points of discontinuity of f in that interval is countable.
Combining this statement with the paragraph above it, yields that these discontinuities must be gaps.
Since any interval (open, half-open and unbouded intervals) can be represented by a countable union of closed intervals, this theorem can be extended to arbitrary intervals since countable unions of countable sets are still countable.
This can be done by using $(a, b) = \bigcup_{n=1}^{\infty}[a + 1/n, b - 1/n]$, or $(a, \infty) = \bigcup_{n=1}^{\infty}[a + 1/n, a + n]$. Similar arguments apply to other types of intervals.

$$
\begin{aligned}
F(\alpha+\gamma)+F(\beta+\gamma) &= F\left[p(\alpha+\beta+\gamma)+(1-p)\gamma\right]+F\left[(1-p)(\alpha+\beta+\gamma)+p\gamma\right] \\
&\leq pF(\alpha+\beta+\gamma)+(1-p)F(\gamma)+(1-p)F(\alpha+\beta+\gamma)+pF(\gamma) \\
&= F(\alpha+\beta+\gamma)+F(\gamma).
\end{aligned}
$$

Part 2 ($\Leftarrow$).
To prove the converse, we will use the continuity of F by lemma 4.1.5. Suppose, $\alpha > \beta$, then

$$
\begin{aligned}
F(\alpha)+F(\beta) &= F\left(\frac{\alpha-\beta}{2}+\frac{\alpha-\beta}{2}+\beta\right)+F(\beta) \\
&\geq 2F\left(\frac{\alpha-\beta}{2}+\beta\right) && \text{(by lemma 4.1.5)} \\
&= 2F\left(\frac{\alpha+\beta}{2}\right),
\end{aligned}
$$

which implies convexity for a continuous F (Hug & Weil, 2020, p. 51, Exercise 6).

□

4.2 Uniqueness-Representation Theorem

Theorem 4.2.1 (Eindeutigkeitssatz zu l^p Metriken)**.**
Suppose $\langle A, \succsim \rangle$ is a complete, segmentally additive, factorial proximity structure, that is also and additive difference structure.
Then there exists a unique $r \geq 1$ and real-valued functions φ_i, defined on A_i, $i = 1, \ldots, n$, that are unique up to a positive linear transformation, such that, for all $a, b, c, d \in A$,

$$(a, b) \succsim (c, d) \iff \delta(a, b) \geq \delta(c, d),$$

where

$$\delta(a, b) = \left[\sum_{i=1}^{n} |\varphi_i(a) - \varphi_i(b)|^r \right]^{1/r}.$$

Proof. This proof proceeds in multiple steps. First, we make observations about the domains and ranges of the functions F_i, use this to find bounds for the domain of F. Using these bounds, we argue that F can be extended to the positive reals. Next, we derive from this a functional equation, that is only satisfied by $f(x) = kx^p$. Finally, we prove that F is equal to this extension on the whole domain, and therefore δ is a Minkowski metric.

$$F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] \tag{4.1}$$

Part 0 (Establishing the basics and using the lemma 4.2.7)**.**
By theorem 4.1.1 and remark 4.1.3, δ must be of the form

$$\delta\left(a,\ b\right) = F^{-1}\left[\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\right]$$

where F is continuous and convex and increasing. By the same argument as in item (iii), δ must be defined on a cartesian product of intervals of the form $[0, \alpha_i)$ where α_i are the ranges of $|\varphi_i(a_i) - \varphi_i(b_i)|$ for $i = 1, \ldots, n$. This cartesian product is convex and, by leaving out 0 in each product, also open. Therefore the requirements for lemma 4.2.7 are met if we leave out $\underline{\mathbf{0}}$ and applying the lemma yields homogeneity for for $\underline{\mathbf{x}}, \underline{\mathbf{z}} \neq \underline{\mathbf{0}}$:

$$\delta\left(\underline{\mathbf{x}},\ \underline{\mathbf{x}} + t(\underline{\mathbf{z}} - \underline{\mathbf{x}})\right) = t\,\delta\left(\underline{\mathbf{x}},\ \underline{\mathbf{z}}\right).$$

Applying homogeneity for the definition of δ above and setting $\alpha_i = |\varphi_i(b_i) - \varphi_i(a_i)|$ yields

$$F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = t \cdot F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right].$$

However, due to translation invariance, homogeneity also holds for $\underline{\mathbf{x}} = \underline{\mathbf{0}}$. [2]

We will show, that the only solutions to this equation are of the form $F(\alpha) = k\alpha^r$, i.e. the power functions.

First, we start by observing the domains and the ranges of F. By theorem 4.1.1, the additive difference model

$$\delta(a,b) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right)$$

yields a metric (that is additive on the coordinate axes) if $F_i(\alpha) = F(t_i\alpha)$ and $F = G^{-1}$ for $i = 1, ..., n$. Thus, F^{-1} is defined for all numbers $\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)$, and each F_i coincides with F on its domain (except for the factor t_i, which scales the argument and adjusts the domain).

Since the domains of F_i are intervals (because δ is a metric with additive segments), we can choose $\omega > 0$ such that for each $i = 1, \ldots, n$ and any α with $0 \leq \alpha \leq \omega/t_i$, α lies within the domain of F_i.

In particular, if we set $\omega_i = \omega/t_i$, then $F^{-1}\left[\sum_{i=1}^{n} F_i(\omega_i)\right] = F^{-1}\left[nF(\omega)\right] =: \Omega$ is defined. Thus, $F = G^{-1}$ is defined at least on the interval $[0, \Omega]$. We can now extend F as follows.

Part 1 (Extending the bounds of F)**.**

For $\alpha < \omega$, we have by eq. (4.1),

$$\begin{aligned} F^{-1}[nF(\alpha)] &= F^{-1}[nF(\alpha\omega/\omega)] \\ &= (\alpha/\omega)F^{-1}[nF(\omega)] \\ &= \alpha\Omega/\omega. \end{aligned} \tag{4.2}$$

Thus F satisfies the following, by plugging in the correct values for α ($\tilde{\alpha} = \alpha\frac{\omega}{\Omega}$ and $\tilde{\alpha} = F^{-1}(\frac{\alpha}{n})$)

$$F(a) = nF(a\omega/\Omega), \quad 0 \leq a \leq \Omega; \tag{4.3}$$

$$F^{-1}(a) = (\Omega/\omega)F^{-1}(a/n), \quad 0 \leq a \leq nF(\omega). \tag{4.4}$$

These two equations can now be used to extend the domain of F and F^{-1}.

We use eq. (4.3) by defining F through the right-hand side (for values where the

[2] By adding $0 = \underline{\mathbf{y}} - \underline{\mathbf{y}}$ for some y in the domain, we get homogeneity:

$$\delta\left(\underline{\mathbf{0}},\ t\underline{\mathbf{z}}\right) = F^{-1}\left[\sum_{i=1}^{n} F(t(y_i - (y_i - z_i)))\right] = tF^{-1}\left[\sum_{i=1}^{n} F(y_i - (y_i - z_i))\right] = t\,\delta\left(\underline{\mathbf{0}},\ \underline{\mathbf{z}}\right).$$

right-hand side is defined, i.e., for $0 \leq a \leq \Omega^2/\omega$). Since $\Omega^2/\omega > \Omega$, this indeed extends F beyond the interval $[0, \Omega]$.
Similarly, F^{-1} satisfies equation eq. (4.4) for the extended interval $0 \leq a \leq nF(\Omega) = n^2 F(\omega)$.
Moreover, eq. (4.1) is now valid for $0 \leq t \leq 1$ and $0 \leq \alpha_i \leq \Omega$:

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} nF(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (4.3))}\\
&= F^{-1}\left[n\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right]\\
&= \frac{\Omega}{\omega}F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i\omega/\Omega)\right] && \text{(by eq. (4.4))}\\
&= \frac{\Omega}{\omega}tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right]\\
& && \text{(by homogeneity applied for } \alpha_i\omega/\Omega \leq \omega)\\
&= tF^{-1}\left[n\sum_{i=1}^{n} F(\alpha_i\omega/\Omega)\right] && \text{(by eq. (4.4))}\\
&= tF^{-1}\left[\sum_{i=1}^{n} nF(\alpha_i\omega/\Omega)\right]\\
&= tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] && \text{(by eq. (4.3))}
\end{aligned}$$

Further, the extended function is both increasing and continuous since the extension is based on scaling of the domain and the range of the original function. For continuity let $a_n \to a$ with $\omega < a < \Omega$. Then

$$\begin{aligned}
\lim_{n\to\infty} F(a_n) &= \lim_{n\to\infty} nF\left(a_n\frac{\omega}{\Omega}\right) && \text{(Definition of } F(t) \text{ for } \omega \leq t \leq \Omega)\\
&= nF\left(\lim_{n\to\infty} a_n\frac{\omega}{\Omega}\right) && \text{(Continuity of } F(t) \text{ for } 0 \leq t \leq \omega)\\
&= nF\left(a\frac{\omega}{\Omega}\right) && \text{(Convergence of } a_n)\\
&= F(a) && \text{(Definition of } F(t) \text{ for } \omega \leq t \leq \Omega)
\end{aligned}$$

which is equivalent to F begin continuous on the extended domain.

For monotonicity we can use a similar argument. Let $a < b$ with $\omega < a, b < \Omega$. Then $a\frac{\omega}{\Omega} < b\frac{\omega}{\Omega}$ holds. Plugging that into F and multiplying by n yields $nF\left(a\frac{\omega}{\Omega}\right) < nF\left(b\frac{\omega}{\Omega}\right)$ due to F being increasing on the original domain. Therefore $a < b$ implies $F(a) < F(b)$ which is equivalent to F increasing. Thus, we have established strict increasing monotonicity, continuity and eq. (4.1) for the extended function.

Repeatedly applying the arguments, extends the intervals in which F is defined and in which monotonicity, continuity and homogeneity (eq. (4.1)) are valid by a factor of Ω/ω each time. Thus, we can extend F to $[0,\infty)$ while preserving these properties. We can prove further, that the homogeneity equation must also hold for $t \in [0,\infty)$. Let $a_i > 0$ and $t > 1$. Then there exists $t' = \frac{1}{t} < 1$ such that $t' \cdot ta_i = a_i$. Therefore

$$\begin{aligned} tt'F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] && (tt'=1) \\ &= F^{-1}\left[\sum_{i=1}^{n} F(tt'\alpha_i)\right] && (tt'=1) \\ &= t'F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] \end{aligned}$$

The last step holds, because both $t' < 1$ and $ta_i < \infty$ hold. Dividing both sides by t' yields

$$F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] \quad \text{for } a, t < \infty \tag{4.5}$$

To sum up, we have obtained an extended function which satisfies monotonicity, continuity and homogeneity for $t \in [0,\infty)$. This extended function must coincide with tie original F on at least the interval $[0,\Omega]$ (because this was the domain of F that we deduced, before extending it).

Part 2 (Deriving functional equation)**.**
Next, we derive a functional equation that allows us to use results on quasi-arithmetic homogeneous means (Hardy et al., 1952, p. 68). The theorem states as follows:

Theorem (Uniqueness of homogeneous quasi-arithmetic means, theorem 4.2.6)**.**
Suppose that ϕ is continuous and strictly increasing in the open interval $(0,\infty)$ and that

$$\mathcal{M}_\phi(t\underline{\mathbf{a}},\ \underline{\mathbf{w}}) = t\mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}}) \quad \textit{with} \quad \mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}}) = \phi^{-1}\left(\sum_{i=1}^{n} w_i\phi(a_i)\right)$$

for all positive $\underline{\mathbf{a}}, \underline{\mathbf{w}}, t$. Then $\mathcal{M}_\phi(a,\ \underline{\mathbf{w}}) = \mathcal{M}_r(a)$. (i.e. $\mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}})$ is either a power mean with $\phi(x) = ax^r + b$ and $a, b, r \in \mathbb{R}, r > 0$ or a geometric mean with $\phi(x) = a\log(x)$, $a \in \mathbb{R}$.)

The proof of this statement will be presented in section 4.2.1, here we will apply the theorem for our proof.

Since all requirements, except the functional equation, are already met by our extension, we will derive this equation now.

Firstly, applying eq. (4.4) on the left hand side yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right].$$

Applying eq. (4.4) once more on the right hand side, yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = F^{-1}\left[\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(\alpha_i)\right] = \Omega/\omega tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right].$$

Hence

$$F^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(t\alpha_i)\right] = tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{n} F(\alpha_i)\right]$$

must hold.

According to Theorem 84 (Hardy et al., 1952), the only monotonic functions satisfying this equation for $0 \leq \alpha < \infty$ are functions of the form $k\alpha^r$ (see also (Aczél, 1966, p. 153)).

Therefore, $F(\alpha) = k\alpha^r$ must hold on at least the interval $[0, \Omega)$.

Problem:

In dem Buch (Hardy et al., 1952) wird der Satz, der hier benutzt wird, so gezeigt, wie ich ihn hier formuliert habe, das heißt insbesondere für alle gewichte $w_i > 0$ mit Summe 1. In In dem Beweis den von Tversky und Krantz (sowohl im Buch als auch im Paper), wird nur gezeigt, dass die Homogenität für Gewichte $w_i = 1/n$ gilt, so wie ich es oben beschrieen habe. Wenn ich die Formulierung auf Wikipedia (link) richtig verstehe, dann reicht dort auch die Homogenität nur für Gewichte $1/n$, jedoch wird auch auf die selben Seiten im Buch von HLP verwiesen, wo man alle positiven Gewichte mit Summe 1 betrachten muss. Jetzt ist mir nicht klar, ob der Fall von den allgemeinen Gewichten direkt aus dem Fall von $1/n$ Gewichten folgt, weil ich sehe das nicht und hab es auch (bisher) nicht geschafft das zu beweisen.

Ich habe auch andersrum versucht, für die Funktion hier herzuleiten, dass Homogenität für die allgemeinen Gewichte gilt, aber habe das auch nicht geschafft. Im Grunde brauche ich dafür die Gleichung eq. (4.2) mit $n = \omega = \lambda$ wobei $\lambda > 0$ gilt, also

$$F^{-1}[\lambda F(\alpha)] = \frac{\alpha}{\lambda}F^{-1}[\lambda F(\lambda)].$$

Wenn ich diese Gleichung hätte, könnte ich daraus die Homogenität für positive Gewichte herleiten, aber leider habe ich es auch nicht geschafft diese Gleichung herzuleiten. Das n kommt ja daher, dass man dort ursprünglich eine Summe hatte und die Homogenität für eine Summe gilt. Aber wenn man versucht das n zu einem λ zu machen, verliert man die Summe... Vielleicht übersehe ich wie man das Problem ganz einfach lösen kann, aber ich stecke hier gerade ziemlich fest. Vielleicht haben Sie ja eine Idee wie man das lösen kann und könnten mir einen Tipp geben, was ich übersehe.

Part 3 (Proving equality for positive reals)**.**
Now, for any $|\varphi_i(a_i) - \varphi_i(b_i)|$, we can find $t > 0$ small enough such that $t|\varphi_i(a_i) - \varphi_i(b_i)| < \omega$ for $i = 1, \ldots, n$. Applying monotonicity yields (we can assume $t \leq 1$):

$$\begin{aligned}
\delta(a,b) &= F^{-1}\left(\sum_{i=1}^{n} F\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}F^{-1}\left(\sum_{i=1}^{n} F\left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}\left(\sum_{i=1}^{n} \left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)^r\right)^{1/r} \qquad (\text{ since } t|\varphi_i(\alpha_i) - \varphi_i(\beta_i)| \leq \omega) \\
&= \left(\sum_{i=1}^{n} |\varphi_i(a_i) - \varphi_i(b_i)|^r\right)^{1/r}
\end{aligned}$$

Thus, δ is a Minkowski metric and, by Theorem Todo: ref add-diff-prox-struct-repr , $F = k \cdot \alpha^r$ and hence $F_i = k(t_i\alpha)^r$ or $F_i(\alpha) = k\alpha^r$ if we incorporate the t_i into the φ_i. Finally, the restriction $1 \leq r < \infty$ follows from the triangle inequality or, more precisely, from Minkowski's inequality.

□

4.2.1 Uniqueness Proof for Homogeneous Means

In this section we will prove the Uniqueness Theorem for Homogeneous means, which states that the power means [3] (including the geometric mean as a limit case) are the only homogeneous quasi-arithmetic means. Before beginning the proof, we will first define these means.

Definition 4.2.2 (Quasi Arithmetic Means and Homogeneity)**.**
Let $\phi : [0, \infty) \to [0, \infty)$ be continuous and strictly increasing, $\underline{\mathbf{a}} = (a_1, \dots, a_n)$ and $\underline{\mathbf{w}} = (w_1, \dots, w_n)$ with $a_i, w_i \geq 0$ for $i = 1, \dots, n$. Then the quasi-arithmetic ϕ-mean of $\underline{\mathbf{a}}$ with weights $\underline{\mathbf{w}}$

$$\mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}}) = \phi^{-1}\left(\sum_{i=1}^{n} w_i \phi(a_i)\right)$$

The mean is said to be *homogeneous*, if

$$\mathcal{M}_\phi(t \cdot \underline{\mathbf{a}},\ \underline{\mathbf{w}}) = t \cdot \mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}})$$

holds for all $t > 0$.

Definition 4.2.3 (Power mean)**.**
For $n \in \mathbb{N}, r > 0$ and $\underline{\mathbf{a}} = (a_1, \dots, a_n)$ with $a_i > 0$ for all $i = 1, ..., n$ we define the power means $\mathcal{M}_r(\underline{\mathbf{a}})$ as

$$\mathcal{M}_r(\underline{\mathbf{a}}) = \left(\frac{1}{n}\sum_{i=1}^{n} a_i^r\right)^{1/r}.$$

For $r = 0$ we can define $\mathcal{M}_0(\underline{\mathbf{a}})$ as the geometric mean

$$\mathcal{M}_0(\underline{\mathbf{a}}) = \sqrt[n]{\prod_{i=1}^{n} a_i}$$

since it is obtained as a limit case for $r \to 0$ (Bullen, 2003, p. 176).

Remark 4.2.4 (Power means are quasi-arithmetic). By setting $\phi(x) = x^r$ for $r > 0$ or $\phi(x) = \log(x)$ we obtain the power means for $r > 0$ and $r = 0$ respectively.

Theorem 4.2.5 (Characterization of equal means)**.**
In order that

$$\mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}}) = \mathcal{M}_\Psi(\underline{\mathbf{a}},\ \underline{\mathbf{w}})$$

for all $\underline{\mathbf{a}}$ *and* $\underline{\mathbf{w}}$*, it is necessary and sufficient that*

$$\phi = \alpha \cdot \psi + \beta$$

where α *and* β *are constants and* $\alpha \neq 0$.

[3] In some literature this is also referred to as the Hölder mean or the generalized means

Theorem 4.2.6 ($\mathcal{M}_r$ are the only homogeneous means $\mathcal{M}_\phi$)**.**
Suppose that ϕ is continuous and strictly increasing in the open interval $(0, \infty)$ and that the mean defined by ϕ is homogeneous, i.e.

$$\mathcal{M}_\phi(t\underline{\mathbf{a}},\ \underline{\mathbf{w}}) = t\mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}})$$

for all positive $\underline{\mathbf{a}}, \underline{\mathbf{w}}$ and t. Then $\mathcal{M}_\phi(\underline{\mathbf{a}},\ \underline{\mathbf{w}})$ is $\mathcal{M}_r(\underline{\mathbf{a}})$.

Preliminary lemma

Notation:

- γ: parametric curve, $\mathrm{L}(\gamma)$: length of curve, $\mathrm{L}_a^x(\gamma)$: length of subcurve from a to x
- δ_∞: ∞-metric, if no subscript then δ is the additive difference metric
- δ: metric, $\delta\,(a,\ b)$ metric distance between a and b, $\delta\,(a) := \delta\,(0,\ a)$
- $\mathrm{B}^\delta_{(r)} := \{x \in \mathbb{R}^n : \delta\,(x) < r\}$: the open Ball with radius r around 0 wrt. the metric δ,
- $\mathrm{B}^\delta_{[r]} := \{x \in \mathbb{R}^n : \delta\,(x) \leq r\}$: the closed Ball with radius r around 0 wrt. the metric δ,
- $\|x, y\|_\infty := \max_{i\leq n}\{|x_i - y_i|\}$: the maximum metric. We just use ∞ to shorten the notation, e.g. in $\mathrm{B}^\infty_{(r)}$.
- $\mathrm{conv}(A)$: convex Hull of A.
- $\mathrm{extreme}(A)$: extreme Points of A.

In this section we will prove the main lemma, which is needed for the proof of the main result theorem 4.2.1. First, we will state the lemma, then give some remarks on the lemma and prove one statement needed for the lemma and then finally prove it. The proof is divided into 5 parts and a whole subsection which is devoted to showing a statement that is needed in the proof. Everything in this section will be assumed as in this Theorem. For example, if we write δ, this implies not any metric but the metric from the lemma with its properties. Moreover all topological terms(open, closed, continuous) are meant with respect to the natural topology of $\mathbb{R}^n$ (i.e. the topology induced by the δ_∞ and the euclidean metric), not the topology induced by δ or the induced topology on S. This will be especially important for the subsection on convexity of the balls. We will begin with the statement:

Lemma 4.2.7 (Main Lemma)**.**
Let S be an open convex subset of n-dimensional Euclidean space. Let δ be a metric on S that satisfies intradimensional subtractivity [4] *with respect to a continuous*

[4]i.e. δ takes the form $\delta\,(x,\ y) = F(|x_1 - y_1|, \ldots, |x_n - y_n|)$.

function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates [5] *in S. If $\langle S, \delta \rangle$ is a metric space with additive segments, then δ is homogeneous, i.e., for any $x, z \in S$ and any real $0 \leq t \leq 1$,*

$$\delta[x, x + t(z - x)] = t\,\delta(x, z).$$

Now we will continue with a remark that makes it possible to move the set such that $0 \in S$ by using the translation invariance of δ.

Remark 4.2.8 (Translation invariance and domain of δ and F).

(i) δ is translation-invariant, i.e. $\delta\,(x + y,\ z + y) = \delta\,(x,\ z)$:

$$\begin{aligned}\delta\,(x + y,\ z + y) &= F[|x_1 + y_1 - (z_1 + y_1)|, \dots, x_n + |y_n - (z_n + y_n)|]\\ &= F[|x_1 - z_1|, \dots, |x_n - z_n|]\\ &= \delta\,(x,\ z)\end{aligned}$$

(ii) Because δ is invariant to translation in its domain, we can move S and redefine the metric on the shifted set. We do this by defining $S' = \{s - x : s \in S\}$ for $x \in S$ and $\delta'(a, b) := \delta\,(a + x,\ b + x)$ for $a, b \in S'$. Since $0 \in S'$, we also can introduce the following shorthand Notation $\delta'(x') := \delta'(0, x')$ for $x' \in S'$. From now on, we will assume that $0 \in S$, because by the argument just provided, we can shift S and δ such that $0 \in S$. Thus we can proceed to proof the statement using the shorthand notation, by which we need to show that:

$$\delta\,(t \cdot z) = t \cdot \delta\,(z) \qquad \forall t \in [0, 1] \text{ and } \forall z \in S'.$$

(iii) Since δ is defined by $\delta\,(x,\ y) = F(|x_1 - y_1|, \dots, |x_n - y_n|)$ and δ is a metric, $F(x) = 0$ iff $x = 0$ must hold. Moreover the function F has to be defined on a set, where at least the "greatest possible difference" in coordinate i is a possible i-th argument. We denote this value as a_i and define it as

$$a_i := \sup\{|x_i - y_i| : (x_1, \dots, x_n), (y_1, \dots, y_n) \in S\} \in \mathbb{R}_{>0} \cup \{\infty\}.$$

Since δ is segmentally additive, every value between 0 and a_i must be in the domain of δ. Since this is true for all i, this implies that F is defined on a cartesian product of half-open, real intervals of the form $[0, a_i)$.

[5] Linear coordinates means that the vector representations are cartesian as opposed to, for example, polar, where each coordinate entry is not the value on the corresponding factor of the cartesian product.

(iv) Moreover, due to the continuity of F and $|\cdot|$, the metric δ is continuous.

Before moving to the proof, we will fist show two statements that we will need right at the start. The first one asserts that the distance between a compact and a closed set is greater zero.

Lemma 4.2.9.
If $I \subset \mathbb{R}^n$ is compact, $Y \subset \mathbb{R}^n$ is closed and I and Y are disjoint, then there exists $\varepsilon > 0$ such that $\delta_\infty(p,\ y) > \varepsilon$ for all $p \in I$ and $y \in Y$.

Proof. We will prove this statement by contradiction. Assume that for all $\varepsilon > 0$ there exists $p \in I$ and $y \in Y$ such that $\delta_\infty(p,\ y) < \varepsilon$. This means, we can construct two sequences $p_n \in I$ and $y_n \in Y$ such that $\delta_\infty(p_n,\ y_n) \to 0$ as $n \to \infty$. I is compact, thus there exists a convergent subsequence p_{n_m} such that $p_{n_m} \to p \in I$ for $n \to \infty$. With the triangle inequality we get

$$\delta_\infty(p,\ y_{n_m}) \leq \delta_\infty(p,\ p_{n_m}) + \delta_\infty(p_{n_m},\ y_{n_m}) \to 0 \text{ as } m \to \infty.$$

Hence, p is a limit point of Y, and since Y is closed, $p \in Y$ must hold which contradicts the assumption. □

The second one allows us to make arbitrary small δ-balls which we will combine with the statement before to assure a distance in δ and not δ_∞.

Lemma 4.2.10 (arbitrary small δ-Balls)**.**
Let $\varepsilon > 0$ such that $\mathrm{B}^\infty_{(\varepsilon)} \subset S$. Then there exists $\alpha > 0$ such that $\mathrm{B}^\delta_{(\alpha)} \subset \mathrm{B}^\infty_{(\varepsilon)}$.

Proof. Since δ is a metric defined by $\delta(x, y) = F[|x_1 - y_1|, \ldots, |x_n - y_n|]$, we can define the j-th partial function of F by setting all other arguments to zero, i.e. $F^j(u_j) := F(0, \ldots, u_j, \ldots, 0)$.

F is strictly increasing in every argument implies that F^j is increasing and the continuity of F directly translates to F^j being continuous. Moreover, due to remark 4.2.8 (iii), $F^j(u_j) = 0$ if and only if $u_j = 0$ holds.

Together, this means that F^j is an increasing, continuous real function with $F^j(0) = 0$.

By setting $\alpha_j = F^j(\varepsilon)$ we get by monotonicity, that $F^j(u_j) < \alpha_j \implies u_j < \varepsilon$. Note, that $F^j(\varepsilon)$ is defined, because $\mathrm{B}^\infty_{(\varepsilon)} \subset S'$ implies that ε is a possible distance for each coordinate. (See also remark 4.2.8 (iii). Using the a_i from the remark, we get that $\varepsilon < a_i$ holds for all i and thus $F^j(\varepsilon)$ is defined for all $j = 1, \ldots n$.)

Doing this for $j = 1, \ldots, n$ yields $\alpha_1, \ldots, \alpha_n$. By setting $\alpha = \min\{\alpha_1, \ldots, \alpha_n\}$ the above holds for every partial function F^j which yields the boundary for the ∞ Metric:

$$F^j(u_j) < \alpha \leq \alpha_j \implies u_j < \varepsilon \quad \forall j = 1, \ldots, n \implies \|u\|_\infty < \varepsilon.$$

Using the monotonicity of F we get

$$\delta\,(u) = F(u) < \alpha \implies F^j(u_j) < \alpha \implies \|u\|_\infty < \varepsilon.$$

$\square$

Now, we will move to the proof of the main lemma. It is suggested to first read the proof to get the main ideas and a big picture understanding, and then read the next section which covers the convexity of spheres which will be of great importance. First we will restate the lemma in the form, in which we will prove it.

Lemma (Restatement of the Lemma)**.**
Let S be an open convex subset of n-dimensional Euclidean space with $0 \in S$. Let δ be a metric on S that satisfies intradimensional subtractivity with with respect to a continuous function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates in S. If $\langle S, \delta \rangle$ is a metric space with additive segments, then δ is homogeneous, i.e., for any $z \in S$ and any real $0 \leq t \leq 1$,

$$\delta\,(tz) = t\,\delta\,(z)\,.$$

The proof consists of 4 parts. First we will find a radius such that all balls with this radius on line from 0 to z are in S. Then we will prove this lemma for $t = \frac{1}{m}$ for $m \in \mathbb{N}$ by proving the two inequalities. After this, we will expand this statement to the rationals and then use continuity to expand it to all real numbers between 0 and 1.

Proof.

Part 0 (Asserting a common distance for all points on the line segment)**.**
Let $z \in S$. Since S is open, $\mathbb{R}^n \backslash S$ is closed. The set $I := \{tz : t \in [0,1]\}$ is compact, since continuous images($t \mapsto tz$) of compact sets ($t \in [0,1]$) are compact. Thus, by lemma 4.2.9 there exists a distance $\varepsilon > 0$ such that $\delta_\infty\,(p,\ y) < \varepsilon$ for all $p \in I$ and $y \in \mathbb{R}^n \setminus S$. This means, for all points on the line from 0 to z the ball with radius ε is a subset of S, i.e. for all $t \in [0,1] : \mathrm{B}^\infty_\varepsilon\,(tz) \subset S$. By lemma 4.2.10, there exists $\alpha > 0$ such that $\mathrm{B}^\delta_{(\alpha)} \subset \mathrm{B}^\infty_{(\alpha)}$, and by translation invariance this holds for all points

on the line: $\mathrm{B}^{\delta}_{\alpha}(tz) \subset S$ for all $t \in [0,1]$. Now let $m \in \mathbb{N}$ such that $\frac{1}{m}\,\delta(z) < \alpha$. In particular, this means that $\mathrm{B}^{\delta}_{(\alpha)}$ and therefore, $\mathrm{B}^{\delta}_{(\frac{1}{m}\,\delta(z))}$ is bounded (with respect to the ∞ metric).

Part 1 ($t\,\delta(z) \le \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
Let $z^{(k)} := \frac{k}{m} \cdot z$ for $k = 0, \dots, m$. $z^{(k)} \in S$ for all k due to convexity of S.

By applying translation invariance, we can calculate the distance between two consecutive points:

$$\delta\left(z^{(k-1)},\ z^{(k)}\right) = \delta\left(\frac{k-1}{m}z,\ \frac{k-1}{m}z + \frac{1}{m}z\right) = \delta\left(\frac{1}{m}z\right) \quad \forall k = 1, \dots, m.$$

Applying the triangle inequality $m-1$-times and using the above result yields

$$\delta(z) \le \sum_{k=0}^{m-1} \delta\left(z^{(k)},\ z^{(k+1)}\right) = m \cdot \delta\left(\frac{1}{m}z\right).$$

Finally, dividing the right side by m yields $\frac{1}{m}\,\delta(z) \le \delta\left(\frac{1}{m}z\right)$, which concludes the first part.

Part 2 ($t\,\delta(z) \ge \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
Since δ is a metric with additive segments, there exists a segment from 0 to z with length $\delta(z)$. remark 3.1.4 yields, that the arc length is a continuous, non-decreasing function. This means, we can divide the segment into m subsegments with length $\frac{1}{m}\,\delta(z)$ and choose $y^{(0)}, \dots, y^{(m)}$ equally spaced on the segment such that $\delta\left(y^{(i)},\ y^{(i-1)}\right) = \frac{1}{m}\,\delta(z)$ and $y^{(0)} = 0$ and $y^{(m)} = z$. Using this, we can define $x^{(i)} := y^{(i)} - y^{(i-1)}$. Translation invariance yields $\delta\left(x^{(i)}\right) = \delta\left(y^{(i)},\ y^{(i-1)}\right) = \frac{1}{m}\,\delta(z)$ and, due to the choice of m in part 0 $x^{(i)} \in S$ holds. Note, that this is still true, if we increase m.

Further, we can observe that $\frac{1}{m}z$ is a convex combination of $x^{(i)}$:

$$\sum_{i=1}^{m} \frac{1}{m}x^{(i)} = \sum_{i=1}^{m} \frac{1}{m}(y^{(1)} - y^{(0)} + y^{(2)} - y^{(1)} + \cdots + y^{(m)}) = \frac{1}{m}(-y^{(0)} + y^{(m)}) = \frac{1}{m}z$$

We showed before, that $\delta\left(x^{(i)}\right) = \frac{1}{m}\,\delta(z)$, which implies $x^{(i)} \in \mathrm{B}^{\delta}_{[\frac{1}{m}\,\delta(z)]}$. By part 0 $\mathrm{B}^{\delta}_{(\frac{1}{m}\,\delta(z))}$ is bounded. Therefore we can apply lemma 4.2.11 from which we get that spheres are convex. This means, that the spheres are closed under convex combination, which implies that $\frac{1}{m}z \in \mathrm{B}^{\delta}_{[\frac{1}{m}\,\delta(z)]}$ must hold. Thus $\delta\left(\frac{1}{m}z\right) \le \frac{1}{m}\,\delta(z)$ which concludes the second part of the proof.

Now, by combining part 1 and part 2, we have established the equality for $t = \frac{1}{m}$. Note, that the equality also holds for all $\mathbb{N} \ni m' > m$ since we only needed m to be big enough in part 2 for $x^{(i)}$ to be in S and for convexity of the spheres. Thus, by making m bigger, the crucial points in part 2 and therefore the equality are still true. Next, we will proceed to extending the equality from $t = \frac{1}{m}$ to $t = \frac{k}{m}$ for $k = 0, \ldots, m$.

Part 3 ($t\,\delta(z) = \delta(tz)$ for $t = \frac{k}{m}$ and $t \in \{0, \ldots, m\}$).
For $z^{(1)}$ we already have $\delta\left(0,\ z^{(1)}\right) = \frac{1}{m}\,\delta(0,\ z)$. Now, for consecutive points, by translation invariance, it holds that

$$\delta\left(z^{(k-1)},\ z^{(k)}\right) = \delta\left(z^{(k-2)},\ z^{(k-1)}\right) = \cdots = \delta\left(z^{(1)},\ z^{(0)}\right) = \delta\left(z^{(1)},\ 0\right) = \frac{1}{m}\,\delta(0,\ z)\,.$$

Using this and the triangle inequality, we get

$$\begin{aligned}
\delta(0,\ z) &\le \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) + \cdots + \delta\left(z^{(m-1)},\ z^{(m)}\right)\\
&= n \cdot \frac{1}{n}\,\delta(0,\ z)\\
&= \delta(0,\ z)\\
\implies \delta(0,\ z) &= \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) + \cdots + \delta\left(z^{(m-1)},\ z^{(m)}\right)
\end{aligned}$$

For the distance $\delta\left(0,\ z^{(2)}\right)$, we get the following two inequalities:

1. $$\delta\left(0,\ z^{(2)}\right) \le \delta\left(0,\ z^{(1)}\right) + \delta\left(z^{(1)},\ z^{(2)}\right) = \frac{2}{m}\,\delta(0,\ z)$$

2. $$\begin{aligned}
\delta(0,\ z) &\le \delta\left(0,\ z^{(2)}\right) + \delta\left(z^{(2)},\ z\right)\\
\delta\left(0,\ z^{(2)}\right) &\ge \delta(0,\ z) - \underbrace{\delta\left(z^{(2)},\ z\right)}_{\le \sum_i \delta\left(z^{(2+i)},\ z^{(2+i+1)}\right)}\\
&\ge \delta(0,\ z) - \sum_{i=0}^{m-3} \delta\left(z^{(2+i)},\ z^{(2+i+1)}\right)\\
&= \delta(0,\ z) - \frac{m-2}{m}\,\delta\left(0,\ z^{(k)}\right)\\
&= \frac{2}{m}\,\delta(0,\ z)
\end{aligned}$$

These two inequalities yield $\delta\left(0,\ z^{(2)}\right) = \frac{2}{m}\,\delta(0,\ z)$. By repeating this for $\delta\left(0,\ z^{(k)}\right)$ for every $k = 2, \ldots, m-1$, we get the equality for every rational number for m big enough. However every rational number with a denominator smaller than m can be represented by another rational number with denominator greater than m. This yields, that the result is true for every rational t.

Part 4 **($t\,\delta(z) = \delta(tz)$ for $t \in [0,1]$).**
The final step can be completed using continuity of the metric. Now let $t \in [0,1]$ and $t_n \in \mathbb{Q}$ with $t_n \to t$ for $n \to \infty$. Then

$$\begin{aligned}
\delta(tz) &= \delta\left(\lim_{n\to\infty} t_n z\right) \\
&= \lim_{n\to\infty} \delta(t_n z) && \text{(continuity)} \\
&= \lim_{n\to\infty} t_n\,\delta(z) && (t_n \in \mathbb{Q} \text{ and part 3}) \\
&= t\,\delta(z)
\end{aligned}$$

□

4.2.2 Convexity of δ Balls

In this subsection we will proof the following lemma which is needed to conclude the proof of lemma 4.2.7. Everything will be assumed as in the lemma 4.2.7. Note, that all topological terms are with respect to the natural topology of the euclidean space.

Lemma 4.2.11 (Convexity of δ-Spheres)**.**
If $\mathrm{B}^{\delta}_{[\alpha]} \subset S$ *is bounded, then it is convex.*

First, we will state some definitions and results on basic convex geometry that we need, and prove the theorem at the end of the section. The definitions and results on convex geometry are taken out of (Hug & Weil, 2020).

First of all, we need the notions of convexity, convex combinations, the convex hull and extreme points. A set is convex if for all two pairs of points of that set, the linear interpolation between these points stays inside the set. By increasing the amount of points, we get what is called the convex combination of these points, which is similar to a linear combination, but has the additional restraint that the coefficients must be positive and add up to 1. The convex hull of a set is, like any other hulls, the smallest convex set that contains the set.

Definition 4.2.12 (Convex Combination and Hull)**.**

1. A set $A \subset \mathbb{R}^n$ is *convex*, iff $(1-a)x + ay \in A$ for all $x, y \in A$ and $a \in [0,1]$.
2. Let $k \in \mathbb{N}$, let $x_1, \dots, x_k \in \mathbb{R}^n$, and let $a_1, \dots, a_k \in [0,1]$ be such that $a_1 + \dots + a_k = 1$. Then $a_1x_1 + \dots + a_kx_k$ is called a *convex combination* of the points $x_1, \dots x_k$.
3. For $A \subset \mathbb{R}^n$ the *convex Hull* $\mathrm{conv}(A)$ is the smallest convex set containing A, i.e.
$$\mathrm{conv}(A) := \cap \{C \subset \mathbb{R}^n : A \subset C, C \text{ convex}\}$$

The next theorem sums up the relationship between convex combination and convex hulls.

Theorem 4.2.13.
For $A \subset \mathbb{R}^n$ *the convex Hull is the set of all convex combination of points in* A*:*

$$\mathrm{conv}(A) = \left\{ \sum_{i=1}^{k} a_i x_i : k \in \mathbb{N}, x_1, \dots, x_k \in A, a_1, \dots, a_k \in [0,1] : \sum_{i=1}^{k} a_i = 1 \right\}$$

Moreover, the convex hull preserves compactness, as the following theorem shows.

Theorem 4.2.14 (convex Hull of compact set is compact).
If $A \subset \mathbb{R}^n$ is compact, then $\operatorname{conv}(A)$ *is compact.*

Next, we will introduce extreme points, which are, in a sense, characteristic points of a set that cannot be represented by a (strict) convex combination.

Definition 4.2.15 (Extreme Point).
Let $A \subset \mathbb{R}^n$ be closed and convex. A point $x \in A$ is called an *extreme point of A* if x cannot be represented as a nontrivial (or strict) convex combination of points of A, that is, if $x = a \cdot y + (1 - a) \cdot z$ with $y, z \in A$ and $a \in (0, 1)$, implies that $x = y = z$. The set of all extreme points of A is denoted by $\operatorname{extreme}(A)$.

For compact and convex sets, the extreme points are enough to "build" the convex hull of the set by doing convex combinations. In particular, this means that for a compact and convex set, every point can be expressed as a convex combination of extreme points.

Theorem 4.2.16 (Minkowski Theorem).
Let $A \subset \mathbb{R}^n$ be compact and convex. Then $A = \operatorname{conv}(\operatorname{extreme} A)$.

Before moving to the convexity of balls, we will first prove two lemma, which we will need while proving convexity of spheres. The first one is the openness/closedness of δ-balls.

Lemma 4.2.17 (δ-Balls are open/closed).
For every radius $r > 0$, the δ-Balls are open $(\mathrm{B}^\delta_{(r)})$ *or closed* $(\mathrm{B}^\delta_{[r]})$.

Proof. $\delta : S' \times S' \to \mathbb{R}_{\geq 0}$ is continuous with $S' \subseteq \mathbb{R}^n$ and therefore $y \mapsto \delta\,(y) = \delta\,(0,\ y)$ is continuous.

Consider the closed interval $U' = [0, \alpha] \subset \mathbb{R}_{\geq 0}$. Then

$$\begin{aligned}
\delta^{-1}(U') &= \{y \in S' : \delta\,(y) \in U'\} \\
&= \{y \in S' : \delta\,(0,\ y) \leq \alpha\} \\
&= \mathrm{B}^\delta_{[\alpha]}
\end{aligned}$$

due to topological continuity of δ (which is equivalent to preimages of closed sets being closed) $\mathrm{B}^\delta_{[\alpha]}$ is closed. By translation invariance this holds for closed balls with any center in S.
Moreover, we can obtain the same result for open balls by repeating this proof and replacing the closed interval with an open one and $\leq$ with $<$. $\square$

The second lemma shows, that extreme points of balls get compressed by a factor of $\alpha \in \mathbb{N}$, if the radius of the ball gets shrinked by the same factor.

Lemma 4.2.18 (compressing extreme points)**.**
Let $\alpha \in \mathbb{R}_{>0}$ and $r \in \mathbb{N}_{>0}$. Further let $\mathrm{E}(\alpha) := \mathrm{extreme}(\mathrm{B}^{\delta}_{[\alpha]})$ and $\mathrm{C}[\alpha] := \mathrm{conv}(\mathrm{B}^{\delta}_{[\alpha]})$. Then, $y \in \mathrm{E}(\alpha)$ implies $\frac{1}{r}y \in \mathrm{E}\left(\frac{\alpha}{r}\right)$.

Remark. We use square brackets for $\mathrm{C}[\alpha]$ to denote that this set is closed. For the reasoning, see the first argument in section 4.2.2.

Proof.
We prove this statement by contraposition.

Assume, that $\frac{1}{r}y$ is not an extreme point, i.e. $\frac{1}{r}y \in \mathrm{C}\left[\frac{\alpha}{r}\right] \setminus \mathrm{E}\left(\frac{\alpha}{r}\right)$. Then we can write $\frac{1}{r}y$ as a strict convex combination of elements $w^i \in \mathrm{C}\left[\frac{\alpha}{r}\right]$:

$$\frac{1}{r}y = \sum_{i=1}^{p} \lambda_i w^i \quad \text{with} \quad \sum_{i=1}^{p} \lambda_i = 1, \quad \lambda_i > 0,\ w^i \in \mathrm{C}\left[\frac{\alpha}{r}\right] \text{ and } p \in \mathbb{N}$$

But because $\mathrm{C}\left[\frac{\alpha}{r}\right] = \mathrm{conv}(\mathrm{B}^{\delta}_{[\frac{\alpha}{r}]})$ holds, we can write any element from $\mathrm{C}\left[\frac{\alpha}{r}\right]$ as a convex combination of elements from $\mathrm{B}^{\delta}_{[\frac{\alpha}{r}]}$ and therefore we can assume $w^{(i)} \in \mathrm{B}^{\delta}_{[\frac{\alpha}{r}]}$.

By translation invariance and the triangle inequality we get for all i:

$$\begin{aligned}
\delta\left(rw^{(i)}\right) &\leq \delta\left(0,\ w^{(i)}\right) + \delta\left(w^{(i)},\ rw^{(i)}\right) && \text{triangle inequality}\\
&= \delta\left(w^{(i)}\right) + \delta\left(0,\ (r-1)\cdot w^{(i)}\right) && \text{translation invariance}\\
&\leq \delta\left(w^{(i)}\right) + \delta\left(0,\ w^{(i)}\right) + \delta\left(w^{(i)},\ (r-1)\cdot w^{(i)}\right) && \text{triangle inequality}\\
&= 2\cdot\delta\left(w^{(i)}\right) + \delta\left(0,\ (r-2)\cdot w^{(i)}\right) && \text{translation invariance}\\
&\dots\\
&\leq r\cdot\delta\left(w^{(i)}\right)\\
&= r\cdot\frac{\alpha}{r} = \alpha\\
\implies rw^{(i)} &\in \mathrm{B}^{\delta}_{[\alpha]}
\end{aligned}$$

This yields, that $y = \sum_{i=1}^{p} \lambda_i r w^{(i)}$ is a strict convex combination (because $\lambda_i r \neq 0$) and therefore y is not in $\mathrm{E}(\alpha)$. □

Now we can prove the convexity of δ-balls.

Proof of lemma 4.2.11. Let $\mathrm{E}(\alpha) := \mathrm{extreme}(\mathrm{B}^{\delta}_{[\alpha]})$ and $\mathrm{C}[\alpha] := \mathrm{conv}(\mathrm{B}^{\delta}_{[\alpha]})$. By lemma 4.2.17 $\mathrm{B}^{\delta}_{[\alpha]}$ is closed and by assumption it is also bounded. Therefore, from the Heine-Borel Theorem, it follows that $\mathrm{B}^{\delta}_{[\alpha]}$ is compact.

By definition and theorem 4.2.14 it follows, that $\mathrm{C}[\alpha]$ is compact (and therefore closed, which is the reason why we denote it with square brackets) and convex. Application of the Minkowski Theorem (theorem 4.2.16) yields:

$$\mathrm{C}[\alpha] = \mathrm{conv}(\mathrm{extreme}(\mathrm{C}[\alpha]))$$

Further, it holds that $\mathrm{extreme}(\mathrm{C}[\alpha]) = \mathrm{extreme}(\mathrm{B}^{\delta}_{[\alpha]})$ (By doing convex combinations, no new extreme points can be introduced because these are the points, which cannot be represented by a convex combination). Combining this and the statement above yields:

$$\mathrm{B}^{\delta}_{[\alpha]} \subset \mathrm{C}[\alpha] = \mathrm{conv}(\mathrm{extreme}(\mathrm{B}^{\delta}_{[\alpha]})).$$

Now, if we prove that $\mathrm{conv}(\mathrm{extreme}(\mathrm{B}^{\delta}_{[\alpha]})) \subset \mathrm{B}^{\delta}_{[\alpha]}$, we get that $\mathrm{B}^{\delta}_{[\alpha]} = \mathrm{C}[\alpha]$ which proves the convexity of $\mathrm{B}^{\delta}_{[\alpha]}$. We simplify this statement before proving it.

$$\mathrm{conv}(\underbrace{\mathrm{extreme}(\mathrm{B}^{\delta}_{[\alpha]})}_{\mathrm{E}(\alpha)}) \subset B$$

$$\Updownarrow$$

$$\sum_{i=1}^{p} \lambda_i y^{(i)} \in \mathrm{B}^{\delta}_{[\alpha]}, \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in \mathrm{E}(\alpha) \\ \lambda_i \in \mathbb{R}_{\geq 0}, \sum_{i=1}^{p} \lambda_i = 1 \end{cases} \qquad \text{(Definition)}$$

$$\Uparrow$$

$$\sum_{i=1}^{p} \lambda_i y^{(i)} \in \mathrm{B}^{\delta}_{[\alpha]}, \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in \mathrm{E}(\alpha) \\ \lambda_i \in \mathbb{Q}_{\geq 0}, \sum_{i=1}^{p} \lambda_i = 1 \end{cases}$$

(B closed, $\mathbb{Q}$ dense in $\mathbb{R}$ [6])

$$\Uparrow$$

$$\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}^{\delta}_{[\alpha]}, \quad r \in \mathbb{N}_{>0},\ y^{(i)} \in \mathrm{E}(\alpha) \qquad (r \text{ common denominator})$$

Thus, we will continue proving that for $r \in \mathbb{R}_{>0}$ and $y^{(i)} \in \mathrm{E}(\alpha)$ it holds that $\sum_{i=1}^{r} \frac{1}{r} y^{(i)} \in \mathrm{B}^{\delta}_{[\alpha]}$.

[6] Density implies, that we can express every real number as a limit of rational numbers. B is closed, and therefore all limits of rational coefficients are included, if the statement holds for rational numbers.

For $y^{(i)} \in \mathrm{E}(\alpha)$ it holds by lemma 4.2.18 that $\frac{1}{r}y^{(i)} \in \mathrm{E}\left(\frac{\alpha}{r}\right) \subset \mathrm{B}^{\delta}_{[\frac{\alpha}{r}]}$. This means, that for the metric distance of the convex combination the following holds:

$$\begin{aligned}
\delta\left(0,\ \sum_{i=1}^{r}\frac{1}{r}y^{(i)}\right) &\leq \delta\left(0,\ \frac{1}{r}y^{(1)}\right) + \delta\left(\frac{1}{r},\ \sum_{i=1}^{r}\frac{1}{r}y^{(i)}\right) && \text{(triangle inequality)}\\
&= \delta\left(0,\ \frac{1}{r}y^{(1)}\right) + \delta\left(0,\ \sum_{i=2}^{r}\frac{1}{r}y^{(i)}\right) && \text{(translation invariance)}\\
&\leq \frac{\alpha}{r} + \delta\left(0,\ \sum_{i=2}^{r}\frac{1}{r}y^{(i)}\right) && \text{(translation invariance)}\\
&\dots && \text{(repeat)}\\
&\leq r\frac{\alpha}{r} = \alpha
\end{aligned}$$

This yields that $\sum_{i=1}^{r}\frac{1}{r}y^{(i)} \in \mathrm{B}^{\delta}_{[\alpha]}$ holds which proves the statement. □

Chapter 5

Discussion

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